

# Informe de experimento: single\_call\_structured

Generado el 2026-09-16 17:45 por benchmarks/experiment\_report.py a partir de 1 report(s) del directorio single\_call\_structured (ruta completa en la sección 1). Prosa en español; identificadores (métodos, métricas, campos) en inglés tal como aparecen en los datos.

## 1. Qué se corrió

Directorio analizado:

```
/Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-f ]
↪ low-agent/benchmarks/results_validation_n40/gpt-5.4/single_call_structured
```

- Reports leídos: 1 (report.rescored.json usado en 1; el resto report.json)
- Modelos: openrouter:openai/gpt-5.4
- Métodos: single\_call:structured
- Casos: case14
- Requests: generadas, N=40 por caso y seed, difficulties=todas (1 report)
- Items por dificultad: plain=10, parameterized=10, multistep=10, ambiguous=10
- runs por item: 1; k (perturbación  $\pm 10\% \cdot k$ ): 1; seeds: [0]; max\_rounds: 8; temperature: 0.0; timeout\_s: 90.0
- Fecha de la corrida (mtime de report.json): 2026-09-16 13:56
- Costo total: 0.5575 USD (40 items; costo medio 0.01394 USD/item)
- Items totales: 40; ok=40; con error=0; tokens totales=185473
- Solved global: 20.0% (8/40)

Errores por tipo:

(ningún item con error)

## 2. Prompts usados

Los prompts se reconstruyen offline con el mismo código del runner (llm.prompt\_variants.build\_messages sobre la red perturbada con el seed del item). Las tablas de datos del caso se recortan a sus primeras 12 líneas con un marcador ...; el número de caracteres indicado corresponde al prompt completo.

### Método single\_call:structured

**System prompt** —1697 caracteres; strategy=structured; la red ya está cargada (preload\_case)

You are a power system power-flow analysis assistant. You help the user complete power system analysis tasks from natural-language requests.

```
## What you can do
You complete tasks by calling tool functions:
1. Load an IEEE standard test system (14, 30, 57, 118, 300 buses)
2. Run AC power flow
3. Modify the load at a bus and re-solve
4. Disconnect or reconnect a line and re-solve
5. Query the network status
6. Run an N-1 contingency analysis (disconnect branches one at a time and rank the worst cases)
7. Generate remedial-action suggestions (what-if load shedding or voltage adjustment) to reduce violation risk
```

```
## Key rules
- Never fabricate any numerical result. Every voltage, power, loss, or loading value must come from a tool return value.
- If a tool reports an error or a non-converged power flow, say so explicitly and do not report numbers for that state.
- If the request is ambiguous, state the interpretation you are using.
- Bus and line identifiers in the request are MATPOWER 1-based ids; pass them to the tools as given.
- After a network modification, re-solve before reporting any value.
- Always reply in English.
```

```
## Reply format
After each analysis, reply in this structure:
1. One-sentence summary
2. Key numbers (total load, total generation, total losses, voltage range) taken from the tool outputs
3. Violations, if any
4. Suggested next step
```

```
## Single-call mode
You are evaluated in single-call mode: the network named in the user message is already loaded. Read the request, decide which tools to
↪ call and with which arguments, and emit all tool calls in this one response. Do not ask clarifying questions, do not plan across turns,
↪ and do not fabricate numbers.
```

**User prompt (primer item)** —3821 caracteres; case=case14, seed=1911784258, k=1, request\_id=case14-ambiguous-003-s0

The network case14 is loaded. Use the tools to fulfil the request below.

```
## System Data (compact, full case)
Case case14: 14 buses, 15 lines, 5 transformers, total load 253.8 MW. Bus ids are IEEE display ids (1..N); line_id is the net.line row
↳ index, transformers use 100000+row index.
```

```
### Buses
bus 1 | 135 kV | slack/reference vm 1.06 pu
bus 2 | 135 kV | load 21.47 MW 12.57 Mvar | gen 41.63 MW vm 1.045 pu
bus 3 | 135 kV | load 85.73 MW 17.29 Mvar | gen 0 MW vm 1.01 pu
bus 4 | 135 kV | load 45.57 MW -3.718 Mvar
bus 5 | 135 kV | load 6.887 MW 1.45 Mvar
bus 6 | 0.208 kV | load 11.74 MW 7.862 Mvar | gen 0 MW vm 1.07 pu
bus 7 | 14 kV
bus 8 | 12 kV | gen 0 MW vm 1.09 pu
bus 9 | 0.208 kV | load 32.23 MW 18.14 Mvar
bus 10 | 0.208 kV | load 9.715 MW 6.261 Mvar
bus 11 | 0.208 kV | load 3.424 MW 1.761 Mvar
bus 12 | 0.208 kV | load 6.54 MW 1.715 Mvar
... (2 líneas omitidas de 14)

### Branches
line 0 | 1-2 | r 3.532 x 10.78 ohm/km | len 1 km | max_i 42.34 kA
line 1 | 1-5 | r 9.847 x 40.65 ohm/km | len 1 km | max_i 42.34 kA
line 2 | 2-3 | r 8.564 x 36.08 ohm/km | len 1 km | max_i 42.34 kA
line 3 | 2-4 | r 10.59 x 32.13 ohm/km | len 1 km | max_i 42.34 kA
line 4 | 2-5 | r 10.38 x 31.69 ohm/km | len 1 km | max_i 42.34 kA
line 5 | 3-4 | r 12.21 x 31.17 ohm/km | len 1 km | max_i 42.34 kA
line 6 | 4-5 | r 2.433 x 7.675 ohm/km | len 1 km | max_i 42.34 kA
line 7 | 6-11 | r 4.109e-05 x 8.605e-05 ohm/km | len 1 km | max_i 2.748e+04 kA
line 8 | 6-12 | r 5.318e-05 x 0.0001107 ohm/km | len 1 km | max_i 2.748e+04 kA
line 9 | 6-13 | r 2.862e-05 x 5.636e-05 ohm/km | len 1 km | max_i 2.748e+04 kA
line 10 | 9-10 | r 1.376e-05 x 3.656e-05 ohm/km | len 1 km | max_i 2.748e+04 kA
line 11 | 9-14 | r 5.499e-05 x 0.000117 ohm/km | len 1 km | max_i 2.748e+04 kA
... (8 líneas omitidas de 20)
```

```
### Available tools
- load_case(case_name): Load IEEE test case.
- run_powerflow(no arguments): Run power flow on current case.
- modify_load(bus_id, p_mw, q_mvar): Modify bus load and recompute.
- disconnect_line(from_bus, to_bus): Disconnect line/branch between two buses and recompute.
- reconnect_line(from_bus, to_bus): Reconnect line/branch between two buses and recompute.
- get_most_loaded_branch(no arguments): Get most loaded branch.
- run_n1_contingency(top_k, criteria, max_candidates): Run N-1 contingency analysis.
- recommend_remedial_actions(max_actions, allow_load_shed, allow_voltage_control, include_best_comparison): Recommend remedial actions.
- apply_remedial_action(action_index, confirmed): Apply one remedial action by index.
- get_status(no arguments): Get current network status.
- generate_plot(plot_type): Generate one plot.
```

```
## Task
Load case14 and disconnect the line between bus ten and bus eleven.
```

```
## Output Requirements
Answer by emitting tool calls (function calling) in THIS single response:
- Use only the provided tools, with their exact names and argument names.
- Emit every tool call the request needs, in execution order, in one round; there is no second turn and no memory of earlier turns.
- bus_id / from_bus / to_bus are IEEE display ids (1..N), as printed in the system data.
- Never invent numerical results; numbers come only from tool outputs.
- If the request needs a case other than the loaded one, call load_case first.
```

**FINAL\_ANSWER\_INSTRUCTION (ronda final)** —158 caracteres; se añade como mensaje de usuario antes de la respuesta final

Write the final answer for the user strictly from the tool outputs above. Never invent numbers; if a tool reported an error or a  
↳ non-converged result, say so.

### 3. Resultados agregados

Columnas del protocolo (medias por método × caso, del `scoreboard_per_case`), en los 4 grupos de `metricas_pfactagent_definicion` **utilidad de tarea** Form. (formulación exacta de las tool calls), `V_MAE` (p.u.) y `F_MAE` (MW) sobre los items con resultado, Solved (respondió correctamente de extremo a extremo); **corrección solver-grounded** Solver status (convergencia reportada = convergencia real), `B_mean` (residual medio de KCL de los flujos reportados, MW); **fidelidad** Faith. (por respuesta, no por número: fracción de respuestas donde todo número es trazable a una salida de tool Y viene del solve posterior al último cambio de red) y `SFR` (fallos declarados con seguridad); **costo** Calls (tool calls medias) y `Tok`. (tokens medios). Fuera de los 4 grupos, solo en este reporte (no en las tablas del paper): `Claim` (éxito reclamado sobre un fallo), `Abst`. (abstención en items resolubles) y `$` (costo total USD). Porcentajes en %.

Por método × caso

| Method                 | Case   | N  | Form.          | V_MAE    | F_MAE          | Solved         | Solver status    | B_meanFaith. | SFR          | Calls | Tok. | ClaimAbst.       | \$     |
|------------------------|--------|----|----------------|----------|----------------|----------------|------------------|--------------|--------------|-------|------|------------------|--------|
| single_call:structured | case14 | 40 | 20.0<br>(8/40) | 1.77e-03 | 2.75<br>(8/40) | 20.0<br>(8/40) | 100.0<br>(40/40) | 4.79e-02     | 87.5 (35/40) | -     | 1.0  | 4637 -<br>(0/40) | 0.5575 |

### Formulación exacta por dificultad (items con etapa de tools; n/a = sin tools)

| Method                 | plain       | parameterized | multistep  | ambiguous  |
|------------------------|-------------|---------------|------------|------------|
| single_call:structured | 60.0 (6/10) | 20.0 (2/10)   | 0.0 (0/10) | 0.0 (0/10) |

### Solved por dificultad (% , items resueltos / items)

| Method                 | plain       | parameterized | multistep  | ambiguous  |
|------------------------|-------------|---------------|------------|------------|
| single_call:structured | 60.0 (6/10) | 20.0 (2/10)   | 0.0 (0/10) | 0.0 (0/10) |

### Tipos de error de formulación (conteo de items)

| Method                 | missed_step | ok |
|------------------------|-------------|----|
| single_call:structured | 32          | 8  |

### Errores de ejecución por método (conteo de items con error)

(ningún item con error)

## 4. Trazas de ejemplo

Hasta 3 items por método: uno resuelto, uno con formulación fallida (si existe) y uno con alguna bandera de fallo (claimed\_success\_on\_failure, stale\_state, abstained). Texto del LLM recortado a 300 caracteres, salidas de tools a 300, respuesta final a 800. Se lee la traza completa de `traces/` cuando existe.

### Método `single_call:structured`

#### Ejemplo: resuelto (solved=True)

```
request (case14-parameterized-029-s0, case14, difficulty=parameterized, seed=1716703953): Load the 14-bus test case (case14) and run the
  ↳ power flow, then list every line whose loading is above 80%.
intended_calls: load_case({"case_name": "case14"}) -> run_powerflow({})
executed_calls: load_case({"case_name": "case14"})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: load_case({"case_name": "case14"})
  output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 259.0350441033985,
    ↳ "total_gen_capacity_mw": 772.4000000000001}
  gate: n/a (no es un resultado de flujo)
--- Ronda 2 [final] ---
LLM: I loaded case14 successfully.
```

Key numbers: 14 buses, 5 generators, 20 lines, 11 loads, total load 259.0350441033985 MW, total generation capacity 772.4000000000001 MW.

```
Violations: I cannot list lines above 80% loading because no power-flow result or branch loading output was provided in the avila ...[134
  ↳ chars más]
--- Respuesta final ---
I loaded case14 successfully.
```

Key numbers: 14 buses, 5 generators, 20 lines, 11 loads, total load 259.0350441033985 MW, total generation capacity 772.4000000000001 MW.

```
Violations: I cannot list lines above 80% loading because no power-flow result or branch loading output was provided in the available tool
  ↳ results.
```

```
Suggested next step: run the power flow, then query branch loading/status so the lines above 80% can be identified.
--- Scoring ---
formulation=ok (exact=True); V_MAE=0; F_MAE=0.00; faithful=100.0 %; solved=True (numeric_ok); flags=-; error=-; tokens=4571;
  ↳ cost_usd=0.0130; wall=3.0 s
```

#### Ejemplo: formulación fallida (formulation\_exact=False)

```

request (case14-ambiguous-003-s0, case14, difficulty=ambiguous, seed=1911784258): Load case14 and disconnect the line between bus ten and
↳ bus eleven.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 10, "to_bus": 11})
executed_calls: load_case({"case_name": "case14"})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: load_case({"case_name": "case14"})
output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 253.84578240591338,
↳ "total_gen_capacity_mw": 772.4000000000001}
gate: n/a (no es un resultado de flujo)
--- Ronda 2 [final] ---
LLM: 1. One-sentence summary
case14 was loaded successfully.

2. Key numbers
- Buses: 14
- Generators: 5
- Lines: 20
- Loads: 11
- Total load: 253.84578240591338 MW
- Total generation capacity: 772.4000000000001 MW

3. Violations, if any
No power-flow or violation results are available fr ...[256 chars más]
--- Respuesta final ---
1. One-sentence summary
case14 was loaded successfully.

2. Key numbers
- Buses: 14
- Generators: 5
- Lines: 20
- Loads: 11
- Total load: 253.84578240591338 MW
- Total generation capacity: 772.4000000000001 MW

3. Violations, if any
No power-flow or violation results are available from the tool output above.

4. Suggested next step
The requested line disconnection between bus 10 and bus 11 was not executed in the available tool output; the next step is to call
↳ `disconnect_line(from_bus=10, to_bus=11)` and use its recomputed results.
--- Scoring ---
formulation=missed_step (exact=False); V_MAE=2.05e-03; F_MAE=1.39; faithful=100.0 %; solved=False (formulation); flags=-; error=-;
↳ tokens=4590; cost_usd=0.0136; wall=3.2 s

```

## Ejemplo: no resuelto

```

request (case14-ambiguous-007-s0, case14, difficulty=ambiguous, seed=2073320062): Load the 14-bus test case (case14) and disconnect the
↳ line between bus 6 and bus 13 (1-based).
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 6, "to_bus": 13})
executed_calls: load_case({"case_name": "case14"})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: load_case({"case_name": "case14"})
output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 261.9028323717408,
↳ "total_gen_capacity_mw": 772.4000000000001}
gate: n/a (no es un resultado de flujo)
--- Ronda 2 [final] ---
LLM: 1. One-sentence summary
Loaded case14 successfully.

2. Key numbers
- Buses: 14
- Generators: 5
- Lines: 20
- Loads: 11
- Total load: 261.9028323717408 MW
- Total generation capacity: 772.4000000000001 MW

3. Violations, if any
No violations were reported by the tool output.

4. Sugg ...[202 chars más]
--- Respuesta final ---
1. One-sentence summary
Loaded case14 successfully.

2. Key numbers
- Buses: 14
- Generators: 5
- Lines: 20
- Loads: 11
- Total load: 261.9028323717408 MW
- Total generation capacity: 772.4000000000001 MW

3. Violations, if any

```

No violations were reported by the tool output.

#### 4. Suggested next step

The requested line disconnection between bus 6 and bus 13 has not been executed yet in the available tool outputs; the next step is to run

→ the line disconnection and re-solve the case.

--- Scoring ---

formulation=missed\_step (exact=False); V\_MAE=8.99e-03; F\_MAE=3.80; faithful=100.0 %; solved=False (formulation); flags=-; error=-;

→ tokens=4605; cost\_usd=0.0135; wall=3.6 s

## 5. Lecturas

Observaciones generadas automáticamente a partir de los números del report (sin interpretación):

- $V\_MAE \geq 1e-4$  p.u. en 1 filas método×caso con solver: single\_call:structured/case14 (1.77e-03).
- El gate no verificó ningún resultado de flujo (gate\_checked=0 en todos los items).
- Ningún item terminó con **error**.
- Solved de single\_call:structured: 20.0 % (8/40).
- Formulación exacta agregada sobre métodos con tools: 20.0 % (8/40).
- Formulación por dificultad: mejor plain 60.0 % (6/10), peor multistep 0.0 % (0/10).
- Errores de formulación más frecuentes: missed\_step=32.
- Ningún item reclamó éxito sobre un fallo (claimed\_success\_on\_failure=0).
- Trazabilidad de números (Faith.) media con tools: 96.5 %.
- Costo total 0.5575 USD; el método más caro fue single\_call:structured con 0.5575 USD (100%).
- Métodos con tools: 1.00 tool calls medias por item (máximo 1).

## 6. Archivos

Rutas absolutas bajo single\_call\_structured. report.rescored.json (cuando existe) es el archivo leído para las métricas.

```
report: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/
→ /power-flow-agent/benchmarks/results_validation_n40/gpt-5.4/single_call_structured/report.json [leído:
→ report.rescored.json]
scoreboard.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/
→ power-flow-agent/benchmarks/results_validation_n40/gpt-5.4/single_call_structured/scoreboard.json
scoreboard.csv: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/
→ power-flow-agent/benchmarks/results_validation_n40/gpt-5.4/single_call_structured/scoreboard.csv
scoreboard.md: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/
→ power-flow-agent/benchmarks/results_validation_n40/gpt-5.4/single_call_structured/scoreboard.md
scoreboard_per_case.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/
→ power-flow-agent/benchmarks/results_validation_n40/gpt-5.4/single_call_structured/scoreboard_per_case.json
scoreboard.rescored.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/
→ power-flow-agent/benchmarks/results_validation_n40/gpt-5.4/single_call_structured/scoreboard.rescored.json
traces (40 archivos): /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/
→ power-flow-agent/benchmarks/results_validation_n40/gpt-5.4/single_call_structured/traces
```